

Everest Engineering College

Sanepa, Lalitpur

Assignment 2 , 2025

Level : BE Software, 3rd Semester

Subject: Calculus II

Due Date: December 31, 2025

Students are required to give their answers in their own words as far as practicable and copy from others is strictly Prohibited.

Attempt all the Questions:

1. Use the power series to solve the following differential equations.

i. $y'' + y = 0$

ii. $y'' - 4xy' + (4x^2 - 2)y = 0$

iii. $(1 - x^2)y'' + 2y = 0$

2. Solve the differential equations by Frobenius method.

i. $3xy'' + 2y' + y = 0$

ii. $y'' + (x - \frac{1}{2})y = 0$

3. Prove the Rodrigue's formula

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n]$$

for the Legendre's polynomial $P_n(x)$.

4. Prove the following recurrence relations

i. $n P_n(x) = x P'_n(x) - P'_{n-1}(x)$

ii. $(2n + 1)P_n(x) = P'_{n+1}(x) - P'_{n-1}(x)$

5. Express the following polynomials as Legendre polynomials.

i. $2x^2 - 4x + 2$

ii. $4x^3 - 2x^2 - 3x + 8$

iii. $x^3 + x + 1$

6. Prove the followings by using Bessel function.

i. $x J'_\nu(x) = -\nu J_\nu(x) + x J_{\nu-1}(x)$

ii. $\frac{d}{dx} [x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x)$

7. Prove that

i. $J_{3/2}(x) = \frac{1}{x} J_{1/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$

ii. $J_{-3/2}(x) = -\frac{1}{x} J_{-1/2}(x) - J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\cos x}{x} - \sin x \right)$

8. Prove that $[J_{1/2}(x)]^2 + [J_{-1/2}(x)]^2 = \frac{2}{\pi x}$

9. Find the Laplace transform of following functions

a. $e^{2t} + 4t^3 - 2\sin 3t + 3\cos 3t$

- b. $\sin at \cos bt$
- c. $\cos^3 2t$ & $\cosh^2 3t$
- d. $e^{2t} \sin t \sin 2t \sin 3t$
- e. $t e^{-3t} \cos 2t$
- f. $\frac{\cos at - \cos bt}{t}$
- g. $\frac{e^{-at} - e^{-bt}}{t}$
- h. $\int_0^t \frac{1 - \cos t}{t} dt$
- i. $\frac{\sin^2 t}{t}$
- j. $\int_0^t t^2 e^{-t} dt$

10. Find the inverse Laplace transform of the following functions

- a. $\frac{1}{s^2 - 5s + 6}$
- b. $\frac{s-1}{(s-1)^2 + 4}$
- c. $\frac{s+1}{s^2 - 6s + 25}$
- d. $\frac{s}{(s+a)^3}$
- e. $\frac{1}{s^4 + 4a^4}$
- f. $\frac{\pi^5}{s^4(s^2 + \pi^2)}$
- g. $\frac{1}{(s^2 + a^2)^2}$
- h. $\log(1 - \frac{a^2}{s^2})$

11. Find the convolution of

- a. $t * e^t$
- b. $1 * \sin at$

12. Use convolution theorem to find the inverse Laplace transform of

- a. $\frac{1}{s(s^2 + 4)}$
- b. $\frac{w}{s^2(s^2 + w^2)}$
- c. $\frac{1}{(s+1)(s+9)^2}$

"ALL THE BEST."